

VizChitra

Angle 1: Two-Thirds Still Flowing

Refined Story Angle with B1 & B2 Integration

"Data, Otherwise" Exhibition, 2026

70 Years of Ganga Basin Hydrological Data, 1951-2021
1,526 River Segments | Five Drainage Tiers | 362 Named Rivers

Data Sources for This Angle

Primary: A2 Mann-Kendall | A6 Segment Classification | A1 Percent Change
Supporting: B1 Basin-Wide SSA | B2 Segment Minima

Generated: June 2026

ANGLE 1

Two-Thirds Still Flowing

1. Headline

Two-Thirds Still Flowing

The headline holds. The B2 finding (74.6% hitting recent minima) is devastating precisely because the headline sets up the stability expectation first. If the headline led with crisis, the B2 finding would be confirmatory, not revelatory. The power of this angle is the three-beat structure: stability on the surface, rhythm in the middle, and the dropped floor underneath. 'Two-Thirds Still Flowing' is the first beat. It must stay because the second and third beats depend on the visitor believing it before the ground shifts.

2. Core Argument

The argument rests on three findings, each reframing the one before it:

- Finding 1 -- Most of the basin is stable (A2, A6).** Of 1,526 river segments tested with the Mann-Kendall trend test, 993 (65.1%) show no significant long-term trend. Only 517 (33.9%) are drying, and 16 (1.0%) are wetting. Think of it like a neighbourhood of 1,500 houses: roughly 1,000 have had steady water pressure for 70 years. About 500 have seen pressure drop -- 177 of those severely (more than 30% loss). The crisis is real but concentrated in specific corridors (Yamuna, Sai-Gomti belt), not everywhere.
- Finding 2 -- The basin breathes between extremes (B1).** Basin-wide SSA reveals dramatic swings year to year. 1961 was the wettest year on record (SSA +2.233), 2009 the driest (SSA -2.014), and 2019 -- just ten years later -- bounced back to the third-wettest (SSA +1.871). The 2000s were the worst decade (mean SSA -0.850, seven drought years), but the 2010s partially recovered (-0.057). Imagine a patient whose resting heart rate is normal but whose pulse swings wildly under stress -- the average looks fine, but the extremes tell a different story. Even the 'stable' segments ride these swings.
- Finding 3 -- The floor has dropped (B2).** 1,139 of 1,526 segments (74.6%) hit their all-time minimum flow after 1991. The worst single year was 2007, when 228 segments (nearly 1 in 7) touched their lowest point simultaneously. Since almost all 517 drying segments hit their minimum post-1991, at least ~620 of

the 993 stable segments must have too. A useful analogy: a lake whose average depth hasn't changed but whose low-water mark keeps sinking further each summer. The trend is flat, but the worst moments are worse than they used to be.

What this means together:

Two-thirds of rivers show no significant long-term direction. Three-quarters have already been lower than at any prior point in the record. The stability is real, but it is thinner than it looks.

Data sources used:

- **A2 (Mann-Kendall)** -- Distinguishes statistically proven trends from noise.
- **A6 (Classification)** -- Maps the mosaic -- drying, stable, wetting, with severity sub-categories.
- **A1 (Percent Change)** -- Gives human-readable scale (e.g., 'flow dropped 42%').
- **B1 (Basin-Wide SSA)** -- Reveals the temporal rhythm of extreme wet and drought years.
- **B2 (Segment Minima)** -- Exposes hidden vulnerability -- when each river hit its all-time low.

Factfulness risks to watch:

- **Negativity Instinct** -- The 74.6% minimum finding could eclipse the genuine stability of two-thirds of the basin.
- **Gap Instinct** -- Visitors may split rivers into 'stable = safe' vs. 'drying = doomed', missing the overlap.
- **Straight Line Instinct** -- Clustering of minima in the 2000s could be read as permanent collapse rather than a severe trough followed by partial recovery.

Final Visual Encoding Specification

The river map uses a narrative-driven temporal animation (1951-2021) as the default experience. The viewer time-scrubs through 70 years; pausing provides analytical filters. All encodings serve the three-beat arc: abundance, breathing, floor-drop.

Channel	Data	Source File	Behaviour	Purpose
Stroke color (hue)	Per-segment annual SSA	segment_ssa_by_year.csv	Animated per year	Delivers three-beat narrative: abundance (early blue) -> breathing (seasonal shifts) -> floor-drop (late amber clustering)
Stroke width	Tier (1-5)	Segment metadata	Fixed	Geographic anchor -- major vs minor rivers always identifiable
Stroke sat. / opacity	B2 min-year accumulation	segment_min_year.csv	Accumulates permanently	Floor-drop reveal -- segments lock in as they hit their all-time minimum; by 2021 the map shows which rivers have hit bottom
Brief pulse	B2 min-year moment	segment_min_year.csv	One-frame event	Draws attention to the exact moment of minimum -- 2007: 228 segments pulse simultaneously
Year label color	B1 basin-wide SSA	basin_annual_ssa.csv	Animated per year	Basin-wide context -- viewer sees whether the whole basin is in drought or surplus that year
Tooltip	A2/A6 class, A1 %, B2 year	Multiple	On pause + hover	Analytical detail for curious viewers: classification, trend, percent change, minimum year

Channel	Data	Source File	Behaviour	Purpose
Pause filters	A6 classification	segment_classification.csv	On pause only	Isolate drying/stable/wetting after emotional context is built; classification is the verdict, not the opening

3. The Tension

Two layered tensions drive the narrative:

- **Tension 1 -- Expectation vs. reality.** A visitor arrives expecting basin-wide collapse. The data says two-thirds of segments have no significant trend. The crisis is real but concentrated (Yamuna, Sai-Gomti), not everywhere. The visitor must revise: the headline story is wrong.
- **Tension 2 -- Stability vs. vulnerability.** Having recalibrated toward reassurance, the visitor then learns that 74.6% of segments -- including many 'stable' ones -- touched their all-time low after 1991 (B2). Stable does not mean safe. The trend is flat, but the troughs are deeper than they used to be.

Why both matter:

The first tension reassures too quickly; the second unsettles too easily. The visitor must hold both: most of the basin isn't trending downward, yet most has already been lower than ever before. That is not a contradiction -- it is the shape of vulnerability in an oscillating system.

4. Key Data Moments

1. The 65.1% reveal (A6, A2).

When the installation loads, the visitor sees the full basin -- and most of it is neither red nor blue. The visual weight of 'no significant trend' is the first surprise: 993 of 1,526 segments in the neutral warm register. Against this calm field, the 517 drying segments cluster visibly in the Yamuna corridor (86.8% drying, median change -21.3%), the Sai-Gomti belt (Sai: 100% drying, -42.1% median change), and the upper Ganga network. The Son (33 segments, 0% drying, +4.8%) and the Kosi (12.1% drying, 0.0% median change) sit quietly stable in the east. This moment counters the assumption that the entire basin is collapsing.

2. The extreme-year rhythm (B1, SSA animation).

As the animation plays, the year label at the bottom-left shifts color with the basin-wide SSA. When 1961 arrives (SSA +2.233), the label turns deep blue and the river strokes flood cool. When 2009 arrives (SSA -2.014), the label turns deep red and the basin overwhelmingly shifts to amber -- the deepest basin-wide drought in the record. Then 2019 -- ten years later -- and the map floods back toward blue, the third-wettest year on record (SSA +1.871). The whiplash is visceral. Even the stable segments participate in this rhythm, revealing that 'no significant trend' does not mean 'no change.'

3. The floor-drop reveal (B2 accumulation).

This is the emotional pivot. As the animation crosses into the 1990s, segments begin accumulating their B2 marks -- a permanent saturation/opacity shift on the stroke when a segment passes its all-time minimum year. At first, the marks appear sporadically. Through the

2000s, they accelerate. In 2007, 228 segments pulse simultaneously -- nearly one in seven rivers in the basin hit their lowest point in the same year. By 2021, the map is covered: 1,139 of 1,526 segments (74.6%) carry the mark. They appear everywhere: on the drying segments (expected), but also densely across the stable ones (not expected). The floor dropped even where the trend did not.

4. The 2007 convergence (B1 + B2).

The single most powerful data moment occurs when the animation reaches 2007. The year label turns red (B1: SSA -1.473). The per-segment SSA colors shift the basin to widespread amber. And 228 segments pulse simultaneously as they hit their all-time minimum -- more than any other year. Three channels converge on one point in time: basin-wide severity (year label), per-segment severity (stroke color), and historical precedent (B2 accumulation pulse) all peak together.

5. Visual Form & Interaction Concept

Form: Temporal river map with SSA-driven animation, accumulating floor-drop, and year-label basin context.

The default experience is a time-scrub animation from 1951 to 2021. The viewer watches 70 years of hydrological history unfold on the river network. Pausing provides analytical filters (show only drying/stable/wetting segments) and tooltips with classification details. A 'Start over' button resets to 1951.

Stroke color: the river's annual state

Each river segment's stroke color is driven by its per-segment annual SSA (Standardised Streamflow Anomaly). SSA normalises each segment to its own mean and standard deviation, so $SSA = -2$ means 'two standard deviations below this river's own average' regardless of absolute flow. The color uses a diverging scale: deep blue ($SSA \geq +1.5$) through pale/neutral ($SSA \sim 0$) to amber/ochre ($SSA \leq -1.5$). As the animation plays, the entire basin shifts color year by year -- cool blues in wet years, warm ambers in drought years. This single animated channel delivers all three narrative beats: abundance in the early decades, the breathing rhythm of wet-dry oscillation, and the clustering of amber in the 2000s.

Stroke width: geographic anchor

Stroke width is fixed by tier (T1: 6px mainstem to T5: 1px headwater). This never changes during the animation, giving the viewer a stable geographic reference -- thick lines are major rivers, thin lines are tributaries. The viewer can always orient themselves regardless of what the color is doing.

B2 accumulation: the floor-drop reveal

The B2 finding (year each segment hit its all-time minimum) is encoded directly on the river strokes through a permanent saturation/opacity shift. When the animation year passes a segment's B2 minimum year, that segment's stroke gains a visible mark -- a subtle darkening, desaturation, or outline shift that persists for the remainder of the animation. In the exact year of minimum, the segment briefly pulses (a 200ms brightness flare) to draw attention. The effect accumulates: by the early 2000s, segments begin locking in rapidly. In 2007, 228 segments pulse simultaneously -- nearly one in seven. By 2021, the map reveals the full pattern: most rivers have been marked, including many whose SSA color has returned to neutral. The floor dropped even where the trend did not.

Year label: basin-wide context

The current year is displayed prominently at the bottom-left corner. Its color is mapped to the basin-wide SSA (B1) using the same diverging scale as the river strokes: deep red for drought years ($SSA < 0$), neutral for average, deep blue for wet years ($SSA > 0$). This zero-clutter encoding communicates basin-wide state without adding UI elements. During the 2000s, the label stays persistently warm/red for seven consecutive years -- the longest red streak in the record. The year label breathes with the basin.

Tooltip and pause filters: analytical layer

When the viewer pauses, they can hover/tap any segment to see: river name, A6 classification (Drying/Stable/Wetting), A2 trend significance (p-value), A1 percent change (early vs late), and B2 minimum flow year. Filter buttons allow isolating drying, stable, or wetting segments. Classification is the verdict -- it appears on pause, not during animation, so the viewer experiences the temporal story before seeing the statistical summary.

The Three-Beat Narrative Arc

Beat 1 -- Stability on the surface. The animation begins in 1951. The basin glows in cool blues and pale neutrals -- most rivers are near or above their long-term average. The year label is calm. Stroke widths show the familiar geography: thick Ganga mainstem, thinner tributaries. The impression forms: a basin at ease. Two-thirds of these rivers will show no significant long-term trend. The visitor doesn't know that yet, but they're seeing it.

Beat 2 -- The basin breathes. As the animation plays through the decades, the basin pulses. 1961 floods blue ($SSA +2.233$) -- the wettest year on record. The 1970s shift warmer. The 1980s oscillate. Then the 2000s arrive: seven consecutive years of warm/red year labels, the river strokes shifting amber across the basin. 2009 is the deepest red ($SSA -2.014$). Then 2019 snaps back to blue ($SSA +1.871$). The impression updates: the basin isn't still, it's rhythmic. Even stable rivers participate in these swings.

Beat 3 -- The floor has dropped. During the animation, B2 marks have been quietly accumulating. Each time a segment passes its all-time minimum year, it pulses and gains a permanent visual mark. In the 1950s-1980s, these appear sporadically. In the 1990s, they accelerate. In 2007, 228 segments pulse at once. By 2021, the map is covered with marks -- 74.6% of segments have been stamped. The viewer pauses, taps a stable segment, and sees: 'Trend: No significant change. All-time minimum: 2007.' The realization lands: the rivers that show no trend have still been lower than they've ever been. The trend is flat, but the floor has dropped.

The arc completes: the visitor has moved from reassurance ('it's mostly stable') through rhythm ('it breathes between extremes') to unsettlement ('the worst moments are recent and widespread'). This is not the trajectory of panic -- it is the trajectory of a more precise understanding.

6. Factfulness Check

Negativity Instinct (heightened by B2) risk:

The finding that 74.6% of segments hit their minimum post-1991 is a powerful alarm. The Negativity Instinct will seize on it: 'Three-quarters of the basin is at its worst!' This flattens what the number actually means. A segment's all-time minimum may have occurred during a single drought year and the river recovered the following year. A minimum is not a permanent state -- it is a moment. And by definition, in any 70-year record with some variability, some period will contain the minimum, even in a perfectly stationary system.

Design guardrail:

The B2 accumulation is embedded in the animation timeline, not shown upfront. The narrative

structure prevents the minimum finding from becoming the only thing the visitor retains. The visitor watches the basin recover after drought years -- SSA colors return to neutral or blue after 2009, even as B2 marks remain. This reinforces that a minimum is a moment, not a permanent state. The tooltip clarifies: 'A river's lowest recorded year is a single event -- not its permanent condition. Many of these rivers recovered the following year.' The brief pulse at the moment of minimum fades into the permanent mark, visually distinguishing the event from the ongoing state.

Straight Line Instinct (heightened by B2) risk:

The clustering of minima in the 2000s (513 segments) and 2010s (425 segments) could suggest an accelerating downward trend -- each decade worse than the last, heading toward permanent collapse.

Design guardrail:

The B2 accumulation itself counters this instinct: marks appear in every decade, not just the 2000s. The 1950s (66), 1960s (80), 1970s (104), 1980s (132) show a gradual increase even in the early period -- not a sudden recent spike from zero. The 1990s (124) actually dips relative to the 1980s before the 2000s surge. The SSA color animation reinforces this: the 2000s were the worst decade (SSA -0.850), but the 2010s recovered to -0.057 and the 2020s are positive (+0.167). The visitor watches the basin recover after the 2000s trough, seeing that the clustering of minima reflects a specific, exceptionally bad decade -- not an irreversible escalation.

Gap Instinct risk:

Visitors may now create a new binary: 'segments that hit their minimum recently' = bad, 'segments that didn't' = fine. This is just the stable/drying binary with a different cut.

Design guardrail:

The B2 accumulation is continuous -- marks appear gradually throughout the animation, not as a binary reveal at 1991. Segments that hit their minimum in 1987 and 1992 are marked only a few animation-seconds apart, preventing a hard boundary. The tooltip states the limitation clearly: 'A single minimum year may reflect one bad monsoon, not sustained change.'

Urgency Instinct risk:

The three-beat arc -- stability, rhythm, floor-drop -- ends on the most unsettling note. The visitor may leave in a state of alarm that overwhelms the preceding nuance.

Design guardrail:

The 'Start over' button lets the visitor replay the animation from 1951, re-entering the stability and rhythm beats before the floor-drop accumulates again. The installation does not lock the visitor in the third beat. The SSA color animation continues through the 2010s recovery -- the last years show the basin returning toward neutral. The final visual impression is the basin's pulse, not just its floor.

7. The Takeaway Feeling

The visitor walks away with a feeling of informed unease -- deeper and more specific than mere 'recalibrated attention.' The first beat (stability) still functions: it prevents panic, it shows the basin is not collapsing everywhere, it gives the visitor a more accurate map of where crisis actually lives. That reorientation holds.

But the B2 finding adds a layer the original angle lacked. The visitor now knows that stability, in the hydrological sense, is not the same as safety. A river with no long-term trend can still hit a record low. A

basin that 'breathes' can still breathe deeper than it ever has before. The feeling is: 'Most of these rivers are still flowing, and most of them have already been lower than anyone alive has ever seen them. Both of those things are true. The question is whether the floor keeps dropping, or whether what I just saw was the worst of it.'

This is not despair. It is the specific discomfort of understanding a system well enough to know that the reassuring statistic and the alarming one describe the same rivers. The visitor does not leave helpless -- they leave holding a more honest question. The response can still be specific. But the specificity now includes watching the floor, not just the trend.